

# altairsim — Altair Disk Extended BASIC 4.1 on an 8" floppy

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altairsim diskbasic.toml

MEMORY SIZE?
LINEPRINTER? C
HIGHEST DISK NUMBER? 0
HOW MANY FILES?
HOW MANY RANDOM FILES?

37033 BYTES FREE
ALTAIR BASIC REV. 4.1
[DISK EXTENDED VERSION]
COPYRIGHT 1977 BY MITS INC.
OK
```

**Altair BASIC Rev 4.1, Disk Extended Version** (MITS, 1977) runs on an 8" Pertec FD-400 floppy drive behind an 88-DCDD board. The DBL boot PROM at `FF00` loads it. The startup of the machine file is `RUN FF00`. On a real disk Altair, the operator did the same: `EXAMINE FF00`, then `RUN`.

This BASIC has a filesystem: files, a directory, `SAVE` by name, and the `DSKINI` command. The cassette BASIC in `../basic` does not have these.

Press `Ctrl-E` to go back to the monitor at any time. Type `RUN` to continue.

## The startup dialogue

BASIC asks five questions. One question has an answer that you cannot guess.

| Question                            | Answer                                                                                  |
|-------------------------------------|-----------------------------------------------------------------------------------------|
| <code>MEMORY SIZE?</code>           | Return. BASIC uses all of the memory.                                                   |
| <code>LINEPRINTER?</code>           | <code>C</code> . BASIC accepts only <code>C</code> , <code>0</code> or <code>Q</code> . |
| <code>HIGHEST DISK NUMBER?</code>   | <code>0</code> . This machine has one drive in use, and the numbers start at zero.      |
| <code>HOW MANY FILES?</code>        | Return                                                                                  |
| <code>HOW MANY RANDOM FILES?</code> | Return                                                                                  |

**Answer** `LINEPRINTER?` with `C`, `0` or `Q`. If you type anything else, BASIC asks the question again and shows no error. A blank line or an `N` makes the machine look stopped, but it is not.

`C` is the 88-C700 line printer. You can answer `C` when the machine has no printer board. The answer only tells BASIC where `LPRINT` sends its output. BASIC writes nothing until you use `LPRINT`. To make `LPRINT` go somewhere, add an 88-C700 board. The command `altairsim -x 'SHOW MACHINE'` `lineprinter` shows a machine that has one.

MIT's wrote the answers for Rev 4.1 in the *Basic Versions* table.

## Delete a character with `_`

The Backspace key does not work in this BASIC. To delete the last character that you typed, type `_` (underscore). This is how a Teletype did it. For example, `PRINT 5Q_` prints `5`.

## See the programs on the disk

1. Type `MOUNT 0`.
2. Type `FILES`.

BASIC does not read the disk until you mount it.

## The files

| File                            | What it is                                                             |
|---------------------------------|------------------------------------------------------------------------|
| <code>diskbasic.toml</code>     | The machine: <code>base = "default"</code> plus the floppy in drive 0. |
| <code>Disk BASIC 4.1.dsk</code> | The bootable system disk.                                              |

**Copy the disk image before you experiment.** Drive 0 is mounted for reading and writing, as on a real machine. `SAVE` and `DSKINI` both write to it. In a clone, `git checkout` puts the image back. A downloaded copy has no way to undo a change.

**Put quotes around the filename.** The filename has spaces in it. Without quotes, `MOUNT` reads `BASIC` as an option and refuses.

Three drives are empty. To fill one, type `MOUNT dsk0:drive1 "my-data.dsk"` at the monitor. Then raise the answer to `HIGHEST DISK NUMBER?`, so that BASIC can see the drive.